Reverse Engineering Project - REPORT

Part I - Studying the Product

External Features

Physical construct
The device features an ergonomic, lightweight black plastic shell composed of an upper top housing and a bottom structural base. It is a wired USB computer mouse featuring a curved plastic housing, two primary click buttons, a central scroll wheel, and a fixed USB cable. Its compact design supports everyday computer navigation, selection, and scrolling.

Subsystems Used:

Mechanical and structural subsystem:

- Includes the upper and lower housing, button covers, scroll-wheel support, glide feet.
- Protects and support the internal electronics.
- Provides stability and smooth movement on a surface.

Button Input subsystem:

- Consists of left and right- click switches.
- Detects button presses and releases.
- Send electrical signals to the controller IC.

Motion sensing subsystem:

- Detects movement of the mouse across a surface.
- In an optical mouse, an LED illuminates the surface and an optical sensor captures images to calculate movement.

The sensor's location and type should be confirmed from your mouse's underside and PCB.

Scroll input subsystem:

Contains the scroll and its encoder mechanism (rotatory encoder).

Converts wheel rotation into electrical signals.

Allows vertical scrolling and may support a middle-click.

USB communication and power subsystem:

Consists of USB cable connector and associated PCB connections.

Receives electrical power from the computer.

Transfers mouse movement and button data to the computer through the USB interface.

Modes of Operation:

Plug and play USB wired mode: operator under standard continuous 2D Motion Tracking, powered directly by a 5V DC supply delivered over the host USB bus.

Purpose of the product:

The mouse simplifies computer operation by enabling efficient navigation, selection, and control through physical movement and button inputs.

User needs addressed:

Easy Navigation: Enables users to move the cursor around the screen effortlessly.

Accurate Selection: Helps select specific icons, files, buttons, and on-screen elements.

Quick Interaction: Supports clicking, double-clicking, right-clicking, and dragging and dropping.

Convenient Scrolling: Allows users to navigate long documents and webpages using the scroll wheel.

Part II – Product origin to latest inventions

1. Difference in Technology Across Versions

Version 1: Mechanical Ball Mouse: Rubber or metal ball with rollers and encoders. Physical ball rotation was used to detect movement.

Version 2: Early Optical Mouse: LED illumination and optical sensing. Replaced the mechanical ball with optical movement detection.

Version 3: Improved Optical Mouse: Higher-resolution optical sensor and improved controller. Provided smoother, more accurate tracking with fewer moving parts.

Version 4: Laser Mouse: Laser-based illumination and optical sensor. Used laser illumination for tracking across a wider range of surfaces in many models.

Version 5: Wireless Mouse: Radio-frequency or Bluetooth communication. Removed the need for a permanent data cable and introduced battery-powered operation.

Version 6: Modern Gaming / Advanced Mouse: High-resolution sensors, high polling rates, programmable controls. Added faster reporting, customisable buttons, and enhanced tracking features.

- ➔ The evolution from ball-based tracking to optical sensing removed the need for a mechanical ball and reduced maintenance. Later improvements introduced laser tracking, USB connectivity, wireless communication, and higher-performance sensors.

2. Engineering importance of this evolution:

The evolution of computer mouse technology is important because it demonstrates how engineering improves the performance, reliability, usability, and efficiency of everyday devices.

1. Improved Accuracy: The transition from mechanical ball tracking to optical and laser sensing improved cursor tracking and reduced mechanical wear.

2. Reduced Maintenance: Optical mice eliminated the need for a physical tracking ball, reducing dust accumulation and cleaning requirements.

3. Better Ergonomics: Improvements in housing design, button placement, and scroll-wheel integration made mice more comfortable and convenient to use.

4. Enhanced Data Transmission: The development of USB interfaces provided a standard method for supplying power and transferring input data between the mouse and computer.

5. Increased Reliability: Fewer moving parts in optical mice reduced mechanical failures and improved long-term usability.

Part III- Internal components and working flow

1. **Introduction of the product:**
   This tear-down report analyzes a standard HP wired USB optical mouse. The hardware structure relies on a single-sided paper-phenolic (FR-2) printed circuit board identified by board markings E206420 NE1000.

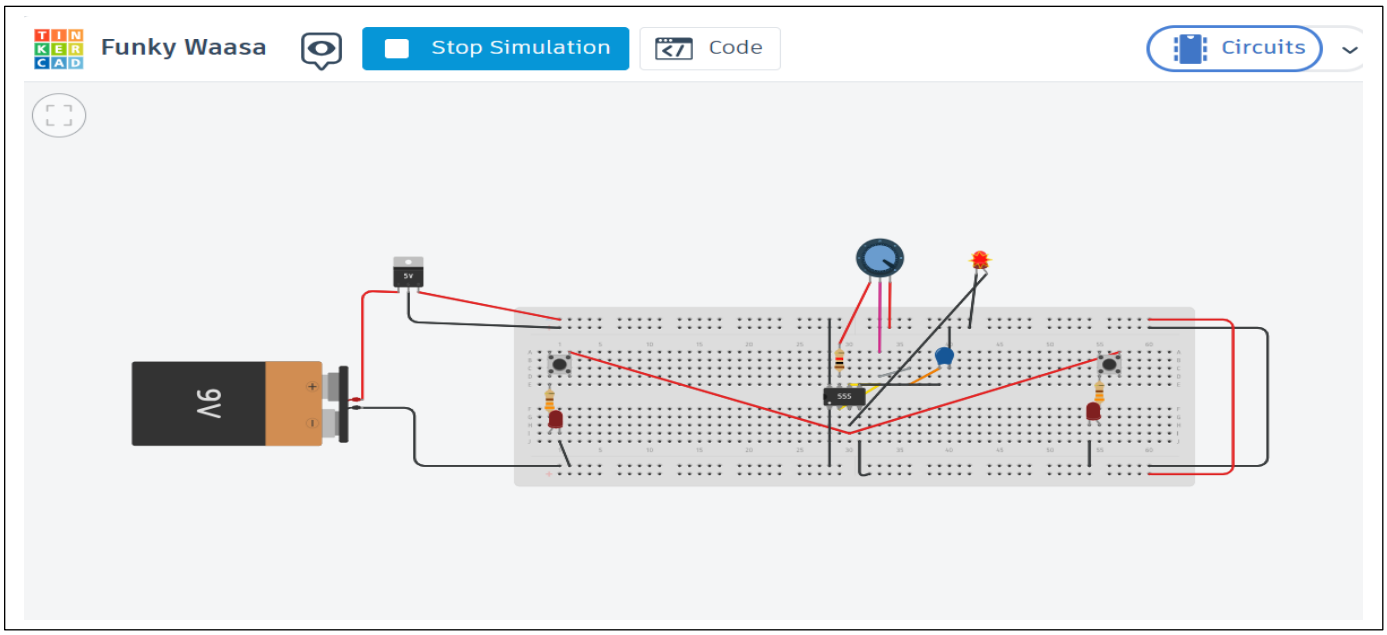
2. **Photograph of the internal components of the product:**

   ->Click on this link or paste it on google for the photos of components and PCB:

   https://drive.google.com/drive/folders/1e4yBBZWSD48wEgcrlH4ZW5WxfdD6DDGS?usp=drive_link

3. **Tinkercad Simulation:**

   You can check the simulation by pasting this link on google : https://www.tinkercad.com/things/8v3IIRBrooQ-funky-waasa



- The bulb is glowing when you press the pushbutton one-by-one.

4. Internal Subsystems with detailed explanation

1. **Input switching subsystem** Main components: Left-click switch, right-click switch, and associated PCB connections.

Purpose: Detects button presses and converts them into electrical signals.

Detailed working:

- When the user presses a mouse button, the plastic button cover pushes a small microswitch.
- The switch contacts change electrical state, usually from open to closed.

- The controller IC reads this change as a digital input.
- When the button is released, the switch returns to its original state.
- The controller communicates the button state to the computer through USB.

2. Motion-sensing subsystem

Main components: Motion sensor, illumination source, and optical lens in an optical mouse.

Purpose: Detects the direction and amount of mouse movement.

Detailed working in an optical mouse:

1. An LED illuminates the surface beneath the mouse.
2. A lens focuses the reflected light onto the optical sensor.
3. The sensor captures successive images of the surface.
4. Its processing circuitry compares these images to determine relative movement along the horizontal and vertical axes.
5. The movement data is sent to the controller, either through a separate interface or as part of an integrated controller-sensor design.

3. Processing and control subsystem

Main components: Controller IC, resistors, capacitors, and any associated timing or interface components.

Purpose: Coordinates the mouse's electronic operation and prepares input data for transmission.

Detailed working:

- The controller reads the left- and right-click signals.
- It processes scroll-encoder inputs and receives motion data from the tracking sensor.
- It may filter unwanted transitions, manage input timing, and combine the inputs into a mouse report.
- Resistors establish appropriate current and signal levels.
- Capacitors help stabilize the power supply and reduce electrical noise.
- Depending on the design, timing may be provided by an internal clock or an external resonator or oscillator.

The controller then sends the formatted data to the USB interface. In some mouse designs, the controller and optical-sensor processing functions are integrated into a single device.

4. USB communication and power subsystem

Main components: USB cable, USB connector, PCB traces, and interface circuitry.

Purpose: Supplies electrical power and transfers mouse data to the computer.

Detailed working:

- The computer supplies nominal 5 V power through the USB connection.
- The power and ground conductors provide the electrical supply and return path.
- The USB data lines carry digital communication between the mouse and computer.
- The mouse identifies itself to the computer through USB enumeration.
- During operation, it sends reports containing button states, movement, and scrolling information.

5. Power conditioning and signal conditioning subsystem

Main components: Resistors, ceramic capacitors, and any additional filtering or protection components fitted to the PCB.

Purpose: Helps maintain suitable electrical conditions for the mouse electronics.

Detailed working:

- Resistors: Limit current, establish logic levels, and may be used in LED or interface circuits.
- Capacitors: Reduce high-frequency power-supply noise and help maintain a stable voltage near the IC.
- Protection components: If present, protect the circuit from certain electrical disturbances or electrostatic discharge.
- PCB traces: Connect the supply, ground, sensor, switches, controller, and USB interface.

Not every mouse has a separate voltage regulator or dedicated protection device; the implementation depends on its circuit design.

COMPLETE WORKING FLOW

Power Delivery: Plugging the USB cable into the host system provides a 5V DC supply to the PCB power rails.

Surface illumination: Resistor-regulated current drives led D1, illuminating the surface beneath the optical aperture.

Image Acquisition: The optical array inside U1 continuously capture microscopic surface profiles at multi-kilohertz frame rates.

Vector calculation: The internal DSP executes surface correlation algorithms to calculate incremental delta X (ΔX) and delta Y (ΔY) movements.

State detection: Button actuation or scroll wheel rotation shifts the digital logic states on the input pins U1.

Data transmission: IC U1 packages the movement vectors and switch states into standardized HID data packets and transmits them over USB data lines (D+, D-) to the host system.

COMPONENTS LIST – BILL OF MATERIALS

Ref. Des.	Component name	Package/ Type	Qty
U1	Optical mouse sensor & IC	DIP-8 Optical Package	1
SW1, SW2	Left/ Right tactile microswitches (TTC)	3- pin through hole	2
BM1	Scroll wheel button switch	2- pin sub miniature	1
ENC1	Rotatory mechanical scroll encoder	Through-hole 8mm height	1
D1	Red / IR illumination led	5mm clear through hole	1
C1, C2, C3, C4, C5	Power decoupling capacitors (ceramic &electrolytic)	Radial lead through-hole	6
R1-R8	Carbon film resistors	Axial through-hole	8

J1	4-pin USB cable header connector	4-pin straight header	1

COMPONENTS FUNCTION & VALUE

S. No.	Component name	PCB Reference	Rating/ value	Function
1	Integrated circuit	U1	8-pin IC socket	Processes signals from the optical sensing system and controls mouse operation
2	Fixed resistor	R1	(O – Orange, B- Black, G- Gold, Br- Brown) OBOG – 30k ohm	Limits current and sets circuit bias
3	Fixed resistor	R2	OBOG- 30k ohm	Signal conditioning and biasing
4	Fixed resistor	R3	OBOG- 30k ohm	Signal conditioning and current control
5	Fixed resistor	R4	OOBG- 33 ohm	Current limiting or biasing
6	Fixed resistor	R5	GGBBr- 550 ohm	Signal conditioning
7	Fixed resistor	R6	GBOO- 33ohm	Signal conditioning
8	Fixed resistor	R7	GBOO- 33 ohm	Signal conditioning
9	Fixed resistor	R8	GBOO- 33 ohm	Biasing or current limiting
10	Jumper / zero-ohm link	JP1	Value not confirmed, not visible properly	Connects two PCB tracks or configures a circuit path
11	Ceramic capacitor	C1	104 = 100nF	Filters noise and stabilizes the supply or signal
12	Ceramic capacitor	C3	104 = 100nF	Signal filtering or timing
13	Ceramic capacitor	C4	‘’	Noise suppression or filtering
14	Ceramic capacitor	C5	‘’	Noise suppression or filtering

15	Electrolytic capacitor	C2	Value is not legible.	Supply filtering and voltage stabilization
16	LED/ optical component	D1	5 mm clear-lens through-hole LED; electrical rating and emitted color not identified.	May serve as an indicator or part of the optical system
17	Microswitch	Left and right buttons	Momentary switch; electrical rating not visible	Detects left- and right-click actions
18	Scroll-wheel encoder	EN1	Mechanical rotary encoder; rating not visible	Detects scroll-wheel rotation
19	Connector	CN1	4-wire mouse cable connector (VCC, GND, data signals)	Connects the PCB to the USB cable
20	USB cable	--------	USB 2.0-compatible interface	Transfers data and supplies power
21	PCB	---------	Marking: E206420, NE1000; 94V-0; thickness marked H=1.6	Supports components and provides electrical connections